

2026 Crossword

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ONE ACROSS

At midnight (12:00 AM) on January 1st, 2026, the hour and minute hands of the world clock coincide. Over the next hour, a time is chosen uniformly at random. Let $\frac{m}{n}$ equal the probability that the straight-line distance between the tips of the hands is less than the clock's radius. What is $m + n$? Assume the hands have equal lengths.

ONE DOWN

As the clock strikes midnight on New Year's Eve, a group of friends decide to exchange party favors. Each friend receives a number of favors equal to the number of divisors of a certain positive integer N . The total number of favors given out is exactly 6. If N is less than 7, what is the sum of all possible values of N ?

TWO DOWN

At a New Year's celebration, 2026 fireworks launch from equally spaced points around a circular lake. Model those launch points as the 2026-th roots of unity on the complex plane. Let $\zeta = e^{2\pi i/2026}$, and define

$$N = \prod_{k=1}^{2025} (2078 - \zeta^k).$$

Determine the last two digits of N .

THREE ACROSS

In a "2026 Countdown Code," the organizers look at the binomial coefficients in row 2026 of Pascal's triangle. Call an integer k with $1 \leq k \leq 2025$ *lucky* if

$$2026 \mid \binom{2026}{k} \text{ but } 2026^2 \nmid \binom{2026}{k}.$$

Suppose the number of lucky integers k is N . Find the largest prime factor of N .